Lecture Notes for Math 372: Elementary Probability and Statistics

Last updated: September 1, 2026

Probability theory has a right and a left hand. On the right is the rigorous foundational work using the tools of measure theory. The left hand "thinks probabilistically," reduces problems to gambling situations, coin-tossing, models of a physical particle. – Leo Breiman

Contents

0 Tentative course outline

This course is a problem-oriented introduction to the basic concepts of probability and statistics, providing a foundation for applications and further study.

- **Weeks 1-3:** Introduction to probability theory
  - Experiments, events, sets, probabilities, random variables. Equally likely outcomes, counting techniques. Conditional probability. Independence. Bayes' theorem. (Sections: 2.1-2.5)
- **Weeks 4-5:** Discrete random variables (1.5 weeks)
  - Expected values, mean, variance, binomial distribution, Poisson distribution. Moment generating functions. (Sections: 3.1-3.6)
- **End of week 5: Midterm 1 (Friday, September 25)**
- **Weeks 6-7:** Continuous Random variables (1.5 weeks)
  - Uniform, exponential, gamma, and normal distributions. Intuitive treatment of the Poisson process and development of the relationship with gamma distributions. (Sections: 4.1-4.4)
- **Weeks 8-9:** Multivariate distributions
  - Calculation of probability, covariance, correlation, marginals, conditions. Distributions of sums of random variables and sampling distributions. Central limit theorem. (Sections: 1.1, 1.3, 1.4, 5.1-5.7)
- **End of week 9: Midterm 2 (Tentatively: Friday, October 23)**
- **Week 10:** Introduction to statistical estimation
  - Point and confidence interval estimation. Maximum likelihood, optimal, and unbiased estimators. Examples. (Sections 6.1, 6.2)
- **Weeks 11-12:** Large sample inference
  - Estimation (1.5 weeks): Types and comparison of estimators; sampling distributions for means/proportions, and their use in large sample estimation; sample size. (Sections 7.1, 7.2)
  - Hypothesis testing (1.5 weeks): Components of a test; signficance and power; p-values; large-sample tests for means and proportions (Sections: 8.1-8.4)
- **Week 13:** Small sample inference
  - t-distribution, with applications to small sample estimation and testing; χ^2 and F distributions, with applications to inference about variances (Sections: 7.3, 7.4, 8.3)
- **Weeks 14-16:** Regression and χ^2 tests
  - Regression (1.5 weeks): Least squares, correlation coefficient, inference (Sections: 12.1-12.5)
  - χ^2 tests: multinomial dsitributions, contingency tables, goodness-of-fit (Sections: 14.1-14.3)
- **Last day of instruction: Wednesday, December 9**
- **Final Exam: December 18**

1 2026-08-24 | Week 01 | Lecture 01

- give syllabus
- did the pirates worksheet

2 2026-08-26 | Week 01 | Lecture 02

> ***The nexus question of this lecture:*** *What is a set and and what are the basic set operations?*
> ***Reading assignment:*** *chapter 2.1*

2.1 Set theory

The mathematical formalization of modern probability theory rests on the foundation of set theory.

2.1.1 Definitions and set-builder notation

- A ***set*** is an unordered collection of distinct objects. The objects of a set are called ***elements***. We write the elements of a set inside curly braces, $\{\ldots\}$.

  Examples: $\{1, 4, 2, 3\}$, $\{\text{cat}, \text{fish}, 5\}$, $\mathbf{Z} := \{\ldots, -2, -1, 0, 1, 2, \ldots\}$, $\mathbf{R} = \{\text{the real numbers}\}$

- Two sets A and B are ***equal***, denoted $A = B$, if they share precisely the same elements. The order that the elements are written is irrelevant.

  $\{1, 2, 3\} = \{3, 1, 2\}$.

  Repeated elements are ignored:

  $\{1, 2, 3\} = \{1, 1, 2, 3\}$.

- A set is ***finite*** if it contains a finite number of distinct elements. This number is called the ***size*** of the set, and is denoted $|A|$.

  Example: $|\{\text{cat}, 5\}| = 2$.

- The ***empty set*** is the set which contains no elements. This is denoted as

  $$\emptyset \quad \text{or} \quad \{\}$$

- ***Set-builder notation*** describes a set by specifying the properties that its elements must satisfy.

  Examples:

  The set of integers between 1 and 36:

  $$\{n \in \mathbb{Z} : 1 \leq n \leq 36\}$$

  The set of ordered pairs (x, y) such that x and y are both positive real numbers:

  $$\left\{(x, y) \in \mathbb{R}^2 : x, y > 0\right\}$$

  The set of even integers:

  $$\{n \in \mathbb{Z} : n \text{ is even}\}$$

  The set of ordered pairs x, y such that both x and y are integers beween 1 and 4:

  $$\{(x, y) : x, y \in \{1, 2, 3, 4\}\}$$

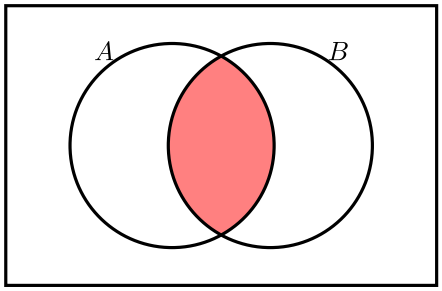
  In set builder notation, the colon ':' is read as "such that" and the symbol '$\in$' (which looks like an E) is read as "is an element of"

2.1.2 Set operations

Let A, B be sets.

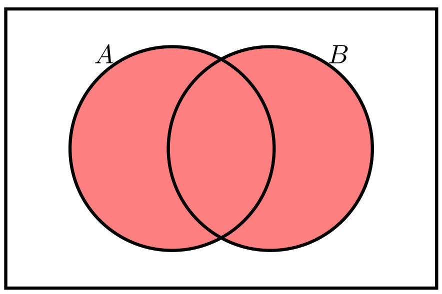
- The ***intersection*** of A and B, denoted $A \cap B$, is defined by

$$A \cap B = \{x : x \in A \text{ and } x \in B\}.$$



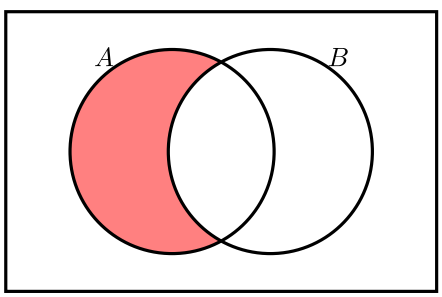
- The ***union*** of A and B, denoted $A \cup B$, is defined by

$$A \cup B = \{x : x \in A \text{ or } x \in B\}.$$



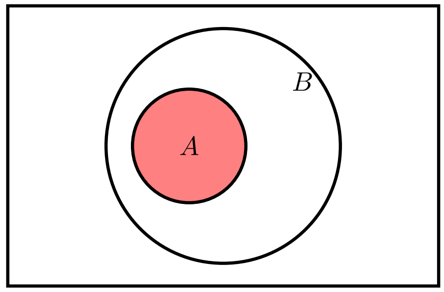
- The ***set difference*** of A and B, denoted $A \setminus B$, is

$$A \setminus B = \{x : x \in A \text{ and } x \notin B\}.$$



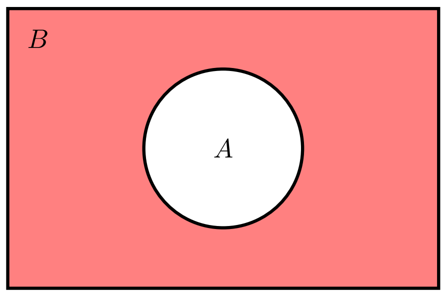
- We say that A is a ***subset*** of B, and write $A \subseteq B$, if

$$x \in A \implies x \in B.$$

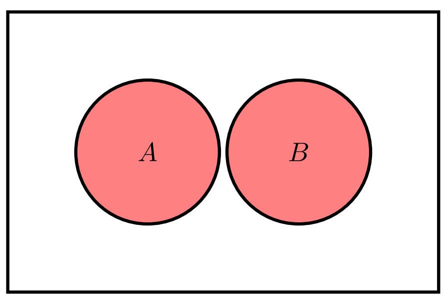


- If $A \subseteq B$, the ***complement of A in B***, denoted A^c, is

$$A^c = B \setminus A = \{x : x \in B \text{ and } x \notin A\}.$$



- We say that A and B are ***disjoint*** if $A \cap B = \emptyset$.



3 2026-08-28 | Week 01 | Lecture 03

> ***The nexus question of this lecture:*** *What does set theory have to do with probability?*
> ***Reading assignment:*** *finish chapter 2.1, start 2.2*

3.1 A general framework for probability

We begin with a general framework and some terminology to formalize the notions of probability.

- An ***experiment*** is an activity or process whose outcome is subject to uncertainty, and about which an observation is made.

  Examples include flipping a coin, rolling a dice, measuring the size of a wave, or the amount of rainfall, conducting a poll, performing a diagnostic test, opening a pack of Pokemon cards, asking your crush on a date, etc.

- The ***sample space*** S of an experiment is the set of all possible outcomes. The elements of the sample space are called ***sample points***. Each sample point represents a **unique** outcome of the experiment.

  Roll a dice. There are six outcomes:

  $$1, 2, 3, 4, 5, 6$$

  These are the sample points. The sample space is

  $$S = \{1, 2, 3, 4, 5, 6\}.$$

- In probability, we refer to subsets of the sample space as ***events***.

  $$\text{an event} = \text{a set of outcomes} = \text{a subset of } S.$$

  We represent events using brackets $[\ldots]$, and usually denote them with capital letters.

  Examples of events when rolling a dice are:

  $$\begin{aligned}
  E_1 &= [\text{roll a 1}] = \{1\} \\
  E_2 &= [\text{roll a 2}] = \{2\} \\
  E_3 &= [\text{roll a 3}] = \{3\} \\
  A &= [\text{roll an odd number}] = \{1, 3, 5\} \\
  B &= [\text{roll an even number}] = \{2, 4, 6\} \\
  C &= [\text{roll a number less than 5}] = \{1, 2, 3, 4\} \\
  D &= [\text{roll a 2 or a 3}] = \{2, 3\}
  \end{aligned}$$

- The first three events shown are ***singleton*** sets, containing only one sample point. The book calls these ***simple events***. But note

  $$1 \neq \{1\}$$

- The empty set $\emptyset$ and the whole sample space S are always both events:

  · S is the event that an outcome occurs ("something happens").

  · $\emptyset$ is the event that no outcome occurs. This is called the ***null event.***

- Two events are E, F ***mutually exclusive*** if $E \cap F = \emptyset$ (i.e., the sets E, F are disjoint). This means that the events E and F cannot both happen at the same time.

  The events A, B defined above are mutually exclusive, since the dice roll cannot be both even and odd.

- Translating set operations into the language of events,

· $E \cap F$ is the event that both E **and** F occur

· $E \cup F$ is the event that at least one of the events E **or** F occurs

· $E^c = S \backslash E$ is the event that E does **not** occur.

Example 1 (Examples of sample spaces)**.**

- Flip a coin. the sample space is $S = \{T, H\}$
- The amount of rainfall in a day is $S = \{x \in \mathbb{R} : x \geq 0\}$

  An example of an event for this sample space is:

$$[\text{between 1 and 2 inches of rain}] = \{x \in \mathbb{R} : 1 \leq x \leq 2\} .$$

- You've got an urn filled with 300,000,000 balls. Exactly one ball is made of gold. Draw a ball out at random. If it's not the gold ball, put it back and keep repeating until you get the gold ball. Once you get the gold ball, you are done. The output of this experiment is *the number of times you drew a ball from the urn.* The sample space is

$$S = \{1, 2, 3, 4, \ldots\} .$$

The last two examples show that the sample space need not be finite.

End of Example 1. □

3.2 Random variable

We often associate numbers with outcomes of an experiment.

Definition 2 (Random variables)**.** Let S be a sample space of an experiment. A ***random variable*** is any rule that associates a number with each outcome in the S. Formally, it is a function whose domain is S and whose range is $\mathbb{R}$.

Most interesting events are defined using random variables.

Example 3 (sum 2d4)**.** Roll a 4-sided dice twice (this is the **experiment**). There are 16 possible **outcomes**. The **sample space** is

$$S = \{(x, y) : x, y \in \{1, 2, 3, 4\}\} ,$$

which consists of 16 ordered pairs (x, y), the **sample points**. We can enumerate the sample space as a table:

		Dice 2			
		1	2	3	4
Dice 1	1	$(1,1)$	$(1,2)$	$(1,3)$	$(1,4)$
	2	$(2,1)$	$(2,2)$	$(2,3)$	$(2,4)$
	3	$(3,1)$	$(3,2)$	$(3,3)$	$(3,4)$
	4	$(4,1)$	$(4,2)$	$(3,4)$	$(4,4)$

Let X be the sum of the two rolls. This is an example of a **random variable**. We can represent X by the following table:

		Dice 2			
		1	2	3	4
Dice 1	1	2	3	4	5
	2	3	4	5	6
	3	4	5	6	7
	4	5	6	7	8

We can define events using X. For example

$$\begin{aligned}
[X \leq 3] &= \{(x,y) \in S : x+y \leq 3\} = \{(1,1),(1,2),(2,1)\} \\
[X = 4] &= \{(x,y) \in S : x+y = 4\} = \{(1,3),(2,2),(3,1)\} \\
[X \leq 4] &= [X \leq 3] \cup [X = 4] = \{(1,1),(1,2),(2,1),(1,3),(2,2),(3,1)\} \\
[X \text{ is even}] &= \{(1,1),(1,3),(2,2),(2,4),(3,1),(3,3),(4,2),(4,4)\}. \\
&\text{etc.}
\end{aligned}$$

End of Example 3. □

3.3 The enumeration principle

On Monday's worksheet, I had you do a bunch of problems where you computed probabilities by counting outcomes. This is a smiple but critically important idea, called the ***enumeration principle***.

> **Enumeration principle:** If the sample space is finite and consists of *equally likely outcomes*, then we can compute lots of probabilities easily by listing outcomes and counting them. To be prcecise, for any event A,
>
> $$\mathbb{P}[A] = \frac{|A|}{|S|} = \frac{\#\text{ of outcomes in } A}{\#\text{ of outcomes in } S}.$$

So using X from Example 3,

$$\mathbb{P}[X = 4] = \frac{3}{16}$$

$$\mathbb{P}[X \text{ is even}] = \frac{8}{16} = \frac{1}{2}.$$

4 2026-08-31 | Week 02 | Lecture 04

The nexus question of this lecture: *What is probability?*
Reading assignment: *finish chapter 2.2, start 2.3*

4.1 Recap and coin flipping example

Recall: The ***sample space*** of an experiment is the *set* of all possible ***outcomes***. An ***event*** is a collection of outcomes. That is,

$$\text{an event} = \text{a set of outcomes} = \text{a subset of the sample space } S.$$

Example 4 (Repeated coin flipping)**.** Consider an experiment in which we flip a coin repeatedly until we get a heads. The output of this experiment is *the number of times we flipped the coin.* There is no upper bound on how many times we might have to flip the coin, so the sample space is

$$S = \{1, 2, 3, 4, \ldots\}.$$

Let $X = (\text{number of coin flips})$. Then

$$\mathbb{P}[X = 1] = \mathbb{P}\left[1^{\text{st}} \text{ flip heads}\right] = \frac{1}{2}$$

and

$$\mathbb{P}[X = 2] = \mathbb{P}\left[1^{\text{st}} \text{ flip tails, } 2^{\text{nd}} \text{ flip heads}\right] = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}.$$

Similarly,

$$\mathbb{P}[X = n] = \left(\frac{1}{2}\right)^n, \qquad n \in \{1, 2, \ldots\}. \tag{1}$$

End of Example 4. □

We will need one more definition.

Definition 5 (Pairwise disjoint)**.** A sequence of events $(A_1, A_2, \ldots)$ is ***pairwise disjoint*** if

$$A_i \cap A_j = \emptyset \quad \text{whenever } i \neq j.$$

In other words, no two events have any "overlap". Equivalently, no two events can simultaneously occur.

4.2 What is probability? The three axioms

We denote the probability of an event E by $\mathbb{P}[E]$ or $\mathbb{P}(E)$. But what is "probability"? And what is $\mathbb{P}$?!

Definition 6 (Probability Axioms)**.** Let S be a sample space associated with an experiment. A function $\mathbb{P}$ is called a ***probability function*** on S if it satisfies the following three axioms:

A.1 (Nonnegativity) $\mathbb{P}[E] \geq 0$ for every event $E \subseteq S$.

A.2 (Sum-to-one) $\mathbb{P}[S] = 1$.

A.3 (Countable additivity) If $A_1, A_2, \ldots$ is an infinite sequence of pairwise disjoint events then

$$\mathbb{P}[A_1 \cup A_2 \cup \ldots] = \mathbb{P}[A_1] + \mathbb{P}[A_2] + \ldots.$$

If $\mathbb{P}$ is a probabilty function, then for every event $E \subseteq S$, the number $\mathbb{P}[E]$ is called the ***probability*** of E.

The above definition may feel unsatisfying becuase it only tells us the conditions that probabilities must satisfy; it doesn't tell us how to assign specific probabilities to events. But before judging this definition, we should first expore the implications of these axioms.

The first two axioms, **A.1** and **A.2** are self-evident. The last one, **A.3**, is more technical and harder to deciper. The next example gives an application of axiom **A.3**.

Example 7 (Continuation of Example 4)**.** What is the probability that we flip the coin an even number of times? That is, what is $\mathbb{P}[X \text{ is even}]$?

Define

$$E_n := [X = n], \qquad n \in \{1, 2, \ldots\}$$

Then

$$[X \text{ is even}] = E_2 \cup E_4 \cup E_6 \cup E_8 \cup \ldots$$

The sequence $E_2, E_4, E_6 \ldots$ is pairwise disjoint. Therefore

$$\begin{aligned}
\mathbb{P}[X \text{ is even}] &= \mathbb{P}[E_2 \cup E_4 \cup E_6 \cup E_8 \cup \ldots] \\
&= \mathbb{P}[E_2] + \mathbb{P}[E_4] + \mathbb{P}[E_6] + \mathbb{P}[E_8] + \ldots && \text{(by } \mathbf{A.3}\text{)} \\
&= \left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^4 + \left(\frac{1}{2}\right)^6 + \left(\frac{1}{2}\right)^8 + \ldots && \text{(by Eq. (1))} \\
&= \frac{1}{4} + \left(\frac{1}{4}\right)^2 + \left(\frac{1}{4}\right)^3 + \left(\frac{1}{4}\right)^4 + \ldots \\
&= \frac{\frac{1}{4}}{1 - \frac{1}{4}} && \text{by geometric series formula } \sum_{n=1}^{\infty} r^n = \frac{r}{1-r} \\
&= \frac{1}{3}.
\end{aligned}$$

So the probability that you flip the coin an *even* number of times is only 1/3.

End of Example 7. □

4.3 Basic properties of probability functions

The next theorem enumerates some basic properties that can be derived using only the three axioms.

Proposition 8 (Basic properties of probabilities)**.**

(i.) *(The probability of no outcome occurring is zero)* $\mathbb{P}[\emptyset] = 0$.

(ii.) *(Finite additivity) If $E_1, \ldots, E_n$ are pairwise disjoint, then*

$$\mathbb{P}[E_1 \cup E_2 \cup \ldots \cup E_n] = \mathbb{P}[E_1] + \mathbb{P}[E_2] + \ldots + \mathbb{P}[E_n].$$

(iii.) *("With probability one, an event E either does occur or doesn't")* $\mathbb{P}[E^c] = 1 - \mathbb{P}[E]$.

(iv.) *(Excision Property) If A, B are events and $A \subseteq B$, then*

$$\mathbb{P}[B \backslash A] = \mathbb{P}[B] - \mathbb{P}[A].$$

"The probability that you get hit by a non-red car is the probability that you get hit by a car minus the probability that you get hit by a red car."

(v.) *("The particular is less likely than the general") If A, B are events and $A \subseteq B$, then $\mathbb{P}[A] \leq \mathbb{P}[B]$*

"The probability that you get hit by a car is higher than the probability that you get hit by a red car."

(vi.) *("Probabilities are between 0 and 1") For any event E, $0 \leq \mathbb{P}[E] \leq 1$.*

Proof of Proposition 8.

- First we will prove (i.). Let $E_i = \emptyset$ for all $i = 1, 2, 3, ...$ Then $(E_i)_{i=1}^{\infty}$ is a pairwise disjoint sequence, and $E_1 \cup E_2 \cup \ldots = \emptyset$. Therefore,

$$\begin{aligned}
\mathbb{P}[\emptyset] &= \mathbb{P}[E_1 \cup E_2 \cup \ldots] \\
&= \mathbb{P}[E_1] + \mathbb{P}[E_2] + \ldots && \text{by } \textbf{A.3} \\
&= \mathbb{P}[\emptyset] + \mathbb{P}[\emptyset] + \ldots.
\end{aligned}$$

  The equation $\mathbb{P}[\emptyset] = (\mathbb{P}[\emptyset] + \mathbb{P}[\emptyset] + \ldots)$ forces $\mathbb{P}[\emptyset] = 0$. This proves (i.).

- Next we prove (ii.). Let $E_1, \ldots, E_n$ be a finite sequence of events which are pairwise disjoint. Then expand the sequence by taking $E_i = \emptyset$ for every positive integer i greater than n. Then

$$\begin{aligned}
\mathbb{P}\left[\bigcup_{i=1}^{n} E_i\right] &= \mathbb{P}\left[\bigcup_{i=1}^{\infty} E_i\right] \\
&= \sum_{i=1}^{\infty} \mathbb{P}[E_i] && \text{by } \textbf{A.3} \\
&= \mathbb{P}[E_1] + \ldots + \mathbb{P}[E_n] + \underbrace{\mathbb{P}[\emptyset] + \mathbb{P}[\emptyset] + \ldots}_{=0 \text{ by (i.)}} \\
&= \mathbb{P}[E_1] + \ldots \mathbb{P}[E_n].
\end{aligned}$$

  This proves (ii.).

- Next we prove (iii.). Let E be any event. Then

$$\begin{aligned}
1 &= \mathbb{P}[S] && \text{by } \textbf{A.2} \\
&= \mathbb{P}[E \cup E^c] && \text{since } S = E \cup E^c \\
&= \mathbb{P}[E] + \mathbb{P}[E^c] && \text{by (ii.), since } E \cap E^c = \emptyset.
\end{aligned}$$

  This proves (iii.).

- Next we prove (iv.). Since $A \subseteq B$,

$$B = A \cup (B \cap A^c).$$

  This is a disjoint union. Therefore by (ii.),

$$\begin{aligned}
\mathbb{P}[B] &= \mathbb{P}[A] + \mathbb{P}[B \cap A^c] \\
&= \mathbb{P}[A] + \mathbb{P}[B \backslash A].
\end{aligned}$$

  Rearranging terms proves (iv.).

- Next we prove (v.). Assume $A \subseteq B$. Then by (iv.),

$$\mathbb{P}[B \backslash A] = \mathbb{P}[B] - \mathbb{P}[A].$$

  Moreover, by **A.1**, $\mathbb{P}[B \backslash A] \geq 0$. Hence

$$0 \leq \mathbb{P}[B] - \mathbb{P}[A]$$

  and this implies (v.).

- Finally, we will show (vi.). Let E be an event. Then

$$\begin{aligned}
0 &\leq \mathbb{P}[E] && \text{by } \textbf{A.1} \\
&\leq \mathbb{P}[S] && \text{by (v.), since } E \subseteq S \\
&= 1 && \text{by } \textbf{A.2}.
\end{aligned}$$

$\square$

5 2026-09-02 | Week 02 | Lecture 05

> ***The topic for this lecture:*** *Combinatorial principles*
> ***Reading assignment:*** *chapter 2.3*

There are three main counting principles we'll look at:

- multiplication principle
- permutation principle
- combination principle

5.1 The multiplication principle

Notation 9. Denote the set consisting of the first n positive integers by

$$[n] = \{1, \ldots, n\}.$$

Definition 10 (Tuple)**.** Let k be a positive integer. A ***k-tuple*** is an ordered list

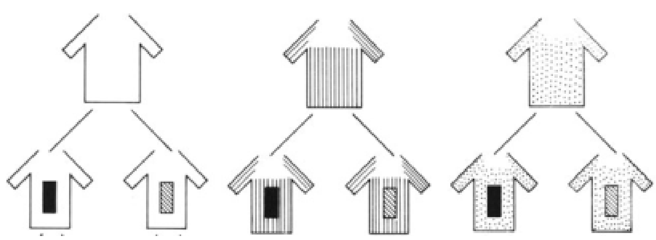
$$(a_1, a_2, \ldots, a_k)$$

of k objects. We call a_i the i^{th} ***coordinate*** of the tuple.

Remark 11. The word *ordered* is key. $(1,2)$ and $(2,1)$ are different 2-tuples, and $(1,1,2)$ is a perfectly fine 3-tuple even thought it repeats an entry.

Theorem 12 (The multiplication principle)**.** *Suppose an object is built by making k choices in order, where the number of options for the 1^{st} choice is n_1, the number of options for the 2^{nd} is n_2, and so forth. Then the number of objects that can be built is $n_1 n_2 \cdots n_k$.*

Example 13. A man has 3 shirts and 2 ties. How many ways can he dress himself?



We can schematize this as three shirts s_1, s_2, s_3 and two ties t_1, t_2, in which case the possible outfits are:

$$\begin{gathered}(s_1, t_1)\ (s_2, t_1)\ (s_3, t_1)\\(s_1, t_2)\ (s_2, t_2)\ (s_3, t_2)\end{gathered}$$

Of course, we don't need to write s and t; it is enough to write

$$\begin{gathered}(1,1)\ (2,1)\ (3,1)\\(1,2)\ (2,2)\ (3,2)\end{gathered}$$

where the first slot is "shirt" and the second is "tie". Equivalently,

$$\{(a_1, a_2) : a_1 \in [3], a_2 \in [2]\}.$$

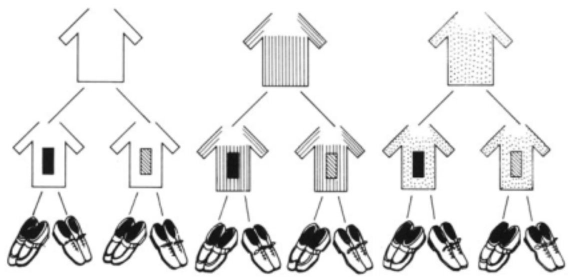
The total number of outfits is $3 \times 2 = 6$.

If the man also has 2 pairs of shoes, we can add a third slot for "shoes", so the question is asking to count the number of elements in the set

$$\{(a_1, a_2, a_3) : a_1 \in [3],\ a_2 \in [2], a_3 \in [2]\}$$

By Theorem 12, the number of elements in this set is $3 \times 2 \times 2 = 12$. Thus the number of ways he can dress himself 12.

This is shown in the following tree diagram:



End of Example 13. □

Corollary 14. $|\{(a_1, a_2, \ldots, a_k) : a_1 \in [n_1],\ a_2 \in [n_2], \ldots, a_k \in [n_k]\}| = n_1 n_2 \cdots n_k$.

Proof. Follows immediately by Theorem 12 □

5.2 Permutations

Example 15. Suppose I have a set of 6 objects:

$$\{\text{coin}, \text{card}, \text{pen}, \text{clip}, \text{craine}, \text{dice}\}$$

Question 1. How many ways can I order these six objects?
Answer: $6! = 6 \times 5 \times 4 \times 3 \times 2 \times 1$.

Question 2. How many ways can I arrange just four elements of S?
Answer: $6! = 6 \times 5 \times 4 \times 3$.

End of Example 15. □

In the remainder, we will formalize these ideas.

Definition 16 (Permutation)**.** Given a set S, a ***permutation*** of S is an ordering of all the elements of S.

Proposition 17 (Counting permutations)**.** *The total number of ways to order n distinct objects is*

$$n! = n \times (n-1) \times (n-2) \times \cdots \times 2 \times 1.$$

(Note that $0! = 1$.)

Definition 18 (Permutation of size k)**.** Given a set S, a ***permutation of size k*** is an ordered subset of S containing exactly k elements.

Theorem 19 (The permutation principle)**.** *Let $P_{k,n}$ be the number of permutations of size k that can be formed from a set of n objects. Then*

$$P_{k,n} = n \cdot (n-1) \cdot (n-2) \cdots (n-k+1) = \frac{n!}{(n-k)!}$$

(Note: the right-hand side is a product of k terms.)

Proof idea. Build the list one entry at a time. There are n choices for the first entry, then $n-1$ for the second (since one element was used up). Then $n-2$, and so on, with $n-k+1$ choices for the k^{th} entry. The number of choices multiply, giving the formula. □

5.3 Combinations

Definition 20 (Combination of size k)**.** Given a set S, a ***combination of size*** k is a subset of S containing exactly k elements.

Theorem 21 (The combination principle)**.** *Let S be a set with n elements. The number of combinations of size k is*

$$\binom{n}{k} = \frac{n!}{k!\,(n-k)!}$$

Proof. Let $C_{k,n}$ be the number of subsets of S of size k. Observe that for every unordered subset of size k, there are $k!$ ways to order it by Proposition 17. Hence,

$$k! \times (\text{number of unordered subsets of size } k) = (\text{number of ordered subsets of size } k).$$

Changing notation,

$$k!\,C_{k,n} = P_{k,n}.$$

Therefore

$$C_{k,n} = \frac{P_{k,n}}{k!}$$

By the formula from Theorem 19,

$$C_{k,n} = \frac{n!}{(n-k)!\,k!}$$

□